CSA6303 - THREAT INTELLIGENCE AND NETWORK SECURITY

Assessment Tool 1: CONCEPT MAPPING

Faculty : Dr.V.Parthipan **SLOT A**

Topics: Network Security Fundamentals | Security Goals (CIA Triad) | Types of Network Threats |Introduction to Threat Intelligence Types of Threat Intelligence (Strategic, Tactical, perational) — Threat Intelligence Lifecycle — Applications in IT Industry

CO1: Interpret security principles, network threats, and the role of threat intelligence in cybersecurity.

Total: 50

Student Name :B.HARI SAI **Register number: 192311098**

Code of Conduct:

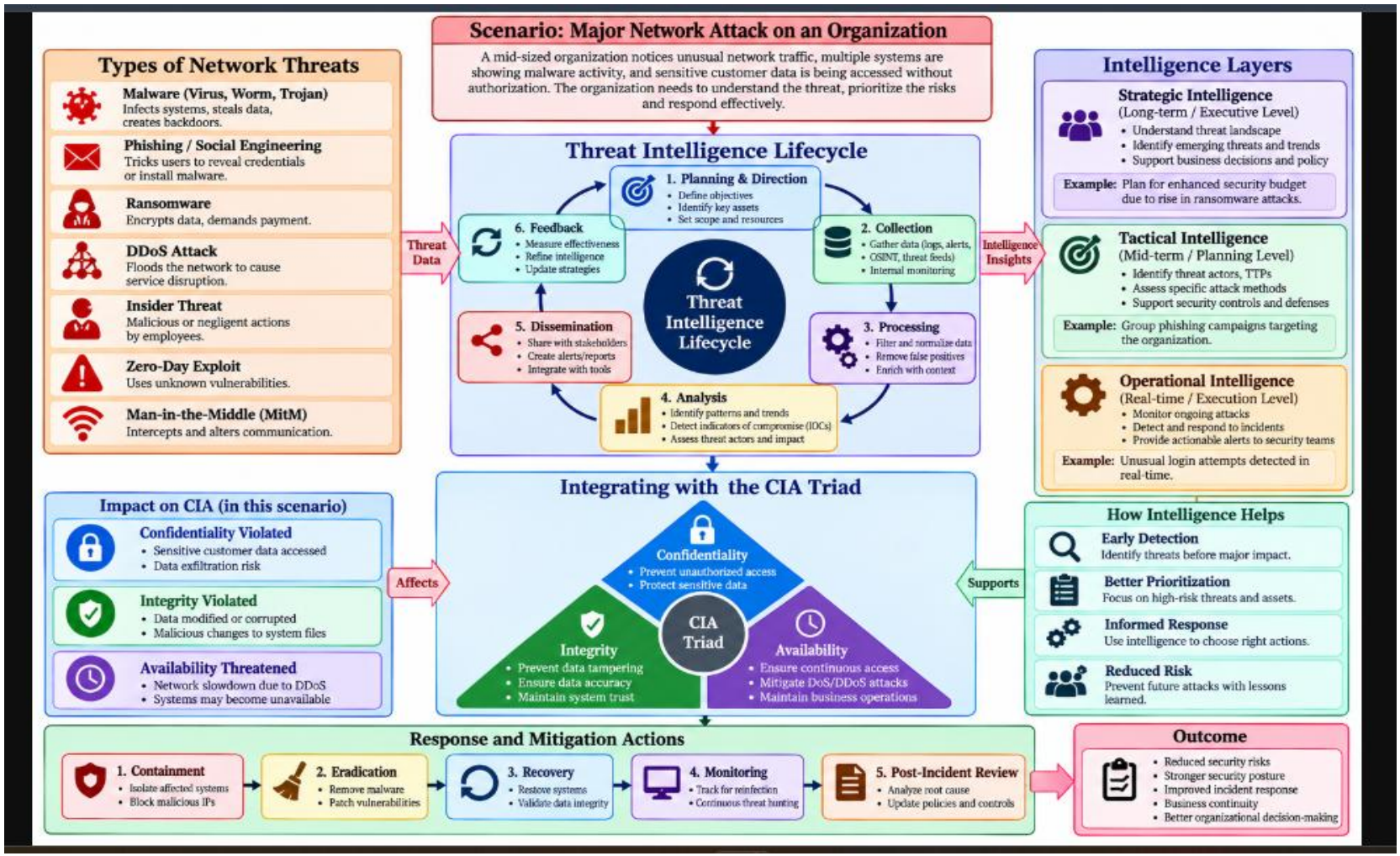
I B.HARI SAI/192311098 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

B.HARI SAI

CO1:CONCEPT MAPPING:

9. Create a scenario-based concept map for an organization experiencing a major network attack. Integrate the CIA Triad, types of network threats, Strategic Intelligence, Tactical Intelligence, Operational Intelligence, and the Threat Intelligence Lifecycle to demonstrate how the organization can understand, prioritize, and respond to the threat.


Scenario: Major Network Attack on an Organization
A mid-sized organization notices unusual network traffic, multiple systems are showing malware activity, and sensitive customer data is being accessed without authorization. The organization needs to understand the threat, prioritize the risks and respond effectively.
Types of Network Threats
Malware (Virus, Worm, Trojan)
Infects systems, steals data, creates backdoors.
Phishing / Social Engineering
Tricks users to reveal credentials or install malware.
Ransomware
Encrypts data, demands payment.
DDoS Attack
Floods the network to cause service disruption.
Insider Threat
Malicious or negligent actions by employees.
Zero-Day Exploit
Uses unknown vulnerabilities.
Man-in-the-Middle (MitM)
Intercepts and alters communication.
Threat Intelligence Lifecycle
1. Planning & Direction
Define objectives
Identify key assets
Set scope and resources
2. Collection
Gather data (logs, alerts, OSINT, threat feeds)
Internal monitoring
3. Processing
Filter and normalize data
Remove false positives
Enrich with context
4. Analysis
Identify patterns and trends
Detect indicators of compromise (IOCs)
Assess threat actors and impact
5. Dissemination
Share with stakeholders
Create alerts/reports
Integrate with tools
6. Feedback
Measure effectiveness
Refine intelligence
Update strategies
Threat Intelligence Lifecycle
Threat Data
Intelligence Insights
Intelligence Layers
Strategic Intelligence
(Long-term / Executive Level)
Understand threat landscape
Identify emerging threats and trends
Support business decisions and policy
Example: Plan for enhanced security budget due to rise in ransomware attacks.
Tactical Intelligence
(Mid-term / Planning Level)
Identify threat actors, TTPs
Assess specific attack methods
Support security controls and defenses
Example: Group phishing campaigns targeting the organization.
Operational Intelligence
(Real-time / Execution Level)
Monitor ongoing attacks
Detect and respond to incidents
Provide actionable alerts to security teams
Example: Unusual login attempts detected in real-time.
Impact on CIA (in this scenario)
Confidentiality Violated
Sensitive customer data accessed
Data exfiltration risk
Integrity Violated
Data modified or corrupted
Malicious changes to system files
Availability Threatened
Network slowdown due to DDoS
Systems may become unavailable
Affects
Integrating with the CIA Triad
Confidentiality
Prevent unauthorized access
Protect sensitive data
Integrity
Prevent data tampering
Ensure data accuracy
Maintain system trust
CIA Triad
Availability
Ensure continuous access
Mitigate DoS/DDoS attacks
Maintain business operations
Supports
How Intelligence Helps
Early Detection
Identify threats before major impact.
Better Prioritization
Focus on high-risk threats and assets.
Informed Response
Use intelligence to choose right actions.
Reduced Risk
Prevent future attacks with lessons learned.
Response and Mitigation Actions
1. Containment
Isolate affected systems
Block malicious IPs
2. Eradication
Remove malware
Patch vulnerabilities
3. Recovery
Restore systems
Validate data integrity
4. Monitoring
Track for reinfection
Continuous threat hunting
5. Post-Incident Review
Analyze root cause
Update policies and controls
Outcome
Reduced security risks
Stronger security posture
Improved incident response
Business continuity
Better organizational decision-making

10.Design a comprehensive concept map demonstrating the applications of Network Security and Threat Intelligence in the IT industry. Connect network security fundamentals, CIA Triad, network threats, Threat Intelligence, Strategic/Tactical/Operational Intelligence, and the Threat Intelligence Lifecycle with applications such as threat monitoring, incident response, vulnerability management, risk assessment, security operations, and organizational decision-making.

